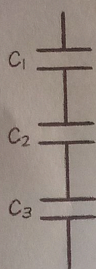
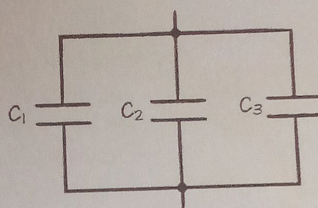


United International UniversityName: Sadika Binti Noor ID: 0112330453Section: X Group: 02 Trimester: 4th Date: 26.01.25**Experiment No. 07****Determination of equivalence capacitance and stored energy by constructing circuits of series and parallel capacitor connections****Theory:** The capacitance of a given capacitor is defined as the ratio of the magnitude of the charge (on either one of the conductors), q , to the voltage (potential difference), V , applied across the two conductors, thus:

$$C = \frac{q}{V} \quad \text{or} \quad C = \frac{dq}{dV}$$



Series connection



Parallel connection

The equivalence capacitance in series can be calculated by the formula

$$C_{\text{total}} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}}$$

The equivalence capacitance in Parallel can be calculated by the formula

$$C_p = C_1 + C_2 + \dots + C_n$$

Capacitance is a measure of the amount of net or excess charge on either one of the conductors per unit potential difference. The more charge a capacitor can store at a given voltage, the larger the capacitance.

The stored energy can be calculated using the formula

$$U = \frac{1}{2} CV^2$$

Here, V is the potential difference between the two parallel plate capacitor.

Apparatus: Voltage source, Capacitors, connecting wires, Voltmeter

Experimental Data:

Table-1:

Series Circuit Connection						
Nominal value of Capacitance (μF)	Measured value of Capacitance (μF)	Measured Equivalent Capacitance, C_{SE} (μF)	Calculated Equivalence Capacitance, C_{ST} (μF)	Applied Voltage, V (Volt)	Measured Stored energy, $U = \frac{1}{2} C_{SE} V^2$ (J)	Calculated Stored energy, $U = \frac{1}{2} C_{ST} V^2$ (J)
1.5×10^{-7}	1.5×10^{-7}	0.64×10^{-7}	0.64×10^{-7}	1.5V	0.72×10^{-3}	0.72×10^{-3}
2.0×10^{-7}	2.0×10^{-7}					
2.5×10^{-7}	2.5×10^{-7}					

Table-2:

Parallel Circuit Connection						
Nominal value of Capacitance (μF)	Measured value of Capacitance (μF)	Measured Equivalent Capacitance, C_{PE} (μF)	Calculated Equivalence Capacitance, C_{PT} (μF)	Applied Voltage, V (Volt)	Measured Stored energy, $U = \frac{1}{2} C_{PE} V^2$ (J)	Calculated Stored energy, $U = \frac{1}{2} C_{PT} V^2$ (J)
1.5×10^{-7}	1.5×10^{-7}	6×10^{-7}	6×10^{-7}	1.5V	6.75×10^{-3}	6.75×10^{-3}
2.0×10^{-7}	2.0×10^{-7}					
2.5×10^{-7}	2.5×10^{-7}					

Calculation: (i) Error and Accuracy for capacitance and energy measurement in series connection:

$$\text{Difference (\%)} \text{ for capacitance} = \frac{\text{Experimental Result} - \text{Theoretical Result}}{\text{Theoretical Result}} \times 100\% = 0\%.$$

$$\text{Accuracy (\%)} = 100\% - \% \text{ Difference} = 100\%.$$

$$\text{and Difference (\%)} \text{ for stored energy} = \frac{\text{Experimental Result} - \text{Theoretical Result}}{\text{Theoretical Result}} \times 100\% = 0\%.$$

$$\text{Accuracy (\%)} = 100\% - \% \text{ Difference} = 100\%.$$

(ii) Error and Accuracy for capacitance and energy measurement in parallel connection:

Physics Laboratory

$$\text{Difference (\%)} \text{ for capacitance} = \frac{\text{Experimental Result} - \text{Theoretical Result}}{\text{Theoretical Result}} \times 100\% = 0\%.$$
$$= 0\%.$$

$$\text{Accuracy (\%)} = 100\% - \% \text{ Difference} = 100\%.$$

$$\text{and Difference (\%)} \text{ for stored energy} = \frac{\text{Experimental Result} - \text{Theoretical Result}}{\text{Theoretical Result}} \times 100\%$$
$$= 0\% \quad = 0\%.$$

$$\text{Accuracy (\%)} = 100\% - \% \text{ Difference} = 100\%.$$

Result:

The equivalence capacitance of the capacitors in series = $0.64 \times 10^{-7} \mu\text{F}$

The equivalence capacitance of the capacitors in Parallel = $6 \times 10^{-7} \mu\text{F}$

Energy stored for capacitors in series = $0.72 \times 10^{-13} \text{ J}$

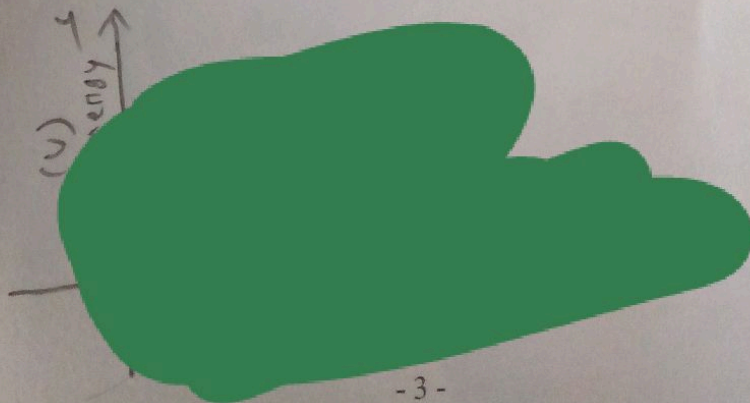
Energy stored for capacitors in parallel = $6.75 \times 10^{-13} \text{ J}$

Discussions:

Q: How can you further increase the stored energy by keeping the dielectric material constant?

$\Rightarrow$ To increase the stored energy in a capacitor while keeping the dielectric material constant, you can increase the capacitance by enlarging the plate area or reducing the distance between the plates, as capacitance is directly proportional to the area and inversely proportional to the plate separation. Additionally, increasing the applied voltage will significantly boost the stored energy since energy is proportional to the square of voltage ($U = \frac{1}{2} CV^2$), but this must remain within the dielectric breakdown limit to avoid failure. Another method is to connect multiple capacitors in parallel, which increase the overall capacitance and energy storage capacity.

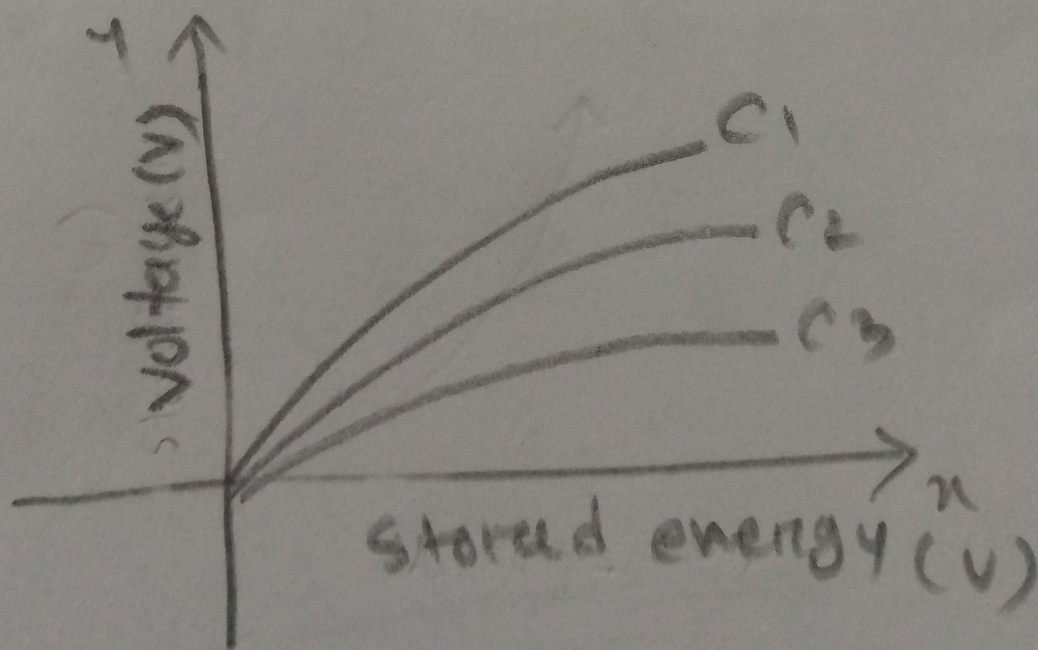
Q: Plot a graph of applied voltage V vs stored energy U for the capacitors with different dielectric materials.



missions:

How can you further increase the stored energy by keeping the dielectric material

increase the stored energy in a capacitor by keeping the dielectric material constant, you can increase the capacitance by enlarging the plate area or decreasing the distance between the plates, as capacitance is proportional to the area and inversely proportional to the separation. Additionally, increasing the voltage significantly boost the stored energy, as energy is proportional to the square of voltage. ($U = \frac{1}{2} CV^2$), but you must stay within the dielectric breakdown limit to avoid failure. To increase energy storage capacity, you can connect multiple capacitors in parallel, which increases the total capacitance. Draw a graph of applied voltage V vs stored energy U for the capacitors with different capacitances.



Q: Do you think the environmental temperature can affect the energy storage process of the capacitor?

=> Yes, environmental temperature can significantly affect a capacitor's energy process. High temperatures can increase leakage currents, reduce the dielectric's breakdown voltage, and decrease capacitance, leading to less efficient energy storage. On the other hand, low temperatures can make the dielectric material brittle, affecting its performance. ~~At~~ Additionally, extreme temperatures may cause physical degradation or thermal expansion of the capacitors are therefore designed to operate within specific temperature ranges for optimal performance.

Q: Write five real life applications where capacitors have been used in energy storage.

- =>
1. Electric vehicles: Capacitors, particularly supercapacitors, are used in EVs to store energy for regenerative braking system, enabling quick energy capture and release.
 2. Uninterruptible Power Supplies: Capacitors are used in UPS system to store energy and provide backup power during short outages, ensuring continuity for critical device.
 3. Renewable Energy system: Capacitors store and smooth energy fluctuations in solar panels and wind turbines.
 4. Electronic Gadgets: Capacitors stabilize power, manage voltage fluctuation, and enable fast charging in phones and laptops.
 5. Power Grids: Capacitors improve power factor, store energy, and enhance grid efficiency in electricity distribution system.